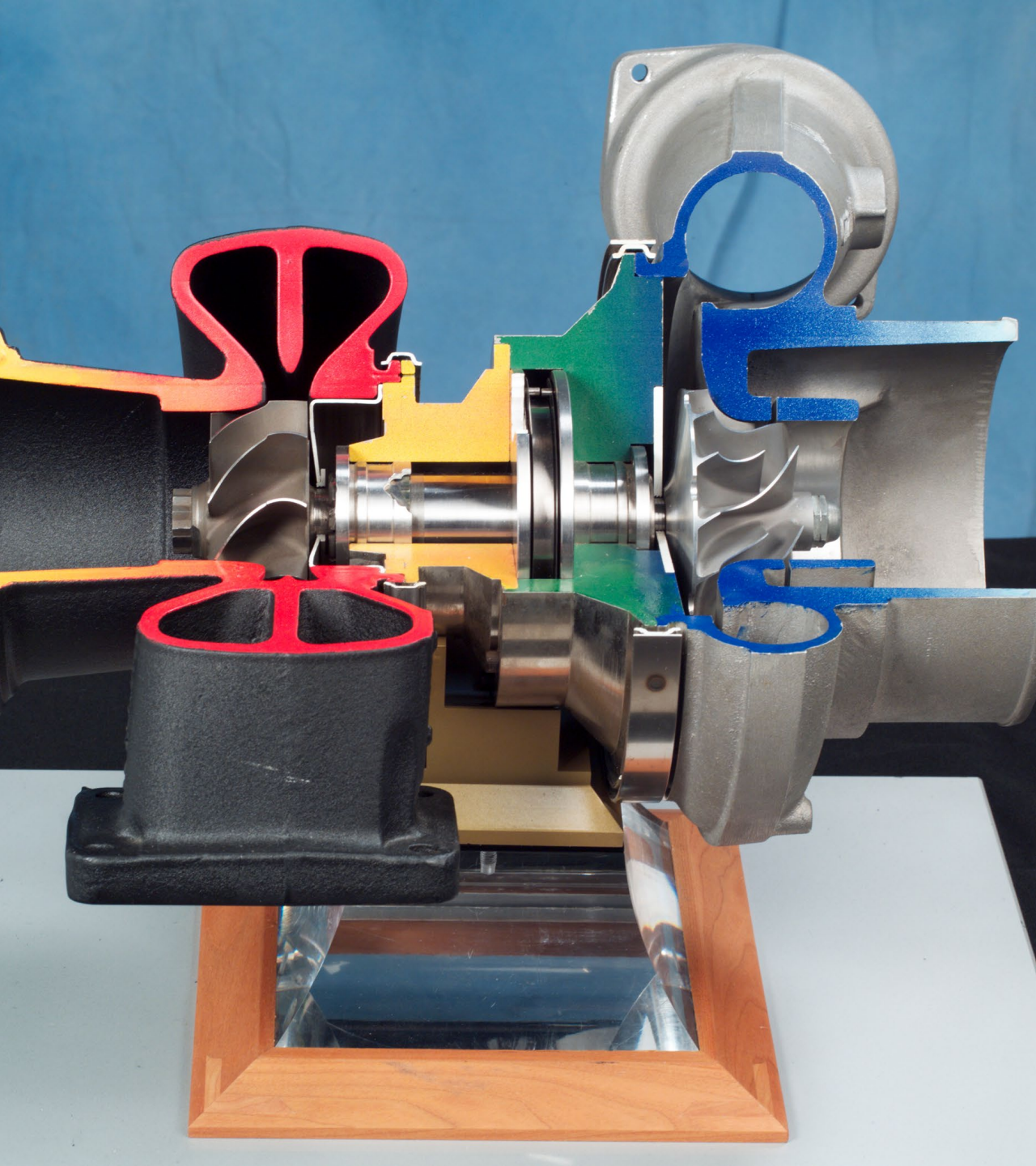




ME 300: Engineering Thermodynamics

Isentropic Efficiency

Isentropic efficiency – turbine

Isentropic efficiency — component efficiency
— compares real comp. to reversible/ideal



Would a non-ideal turbine produce greater than, less than, or the same amount of work as an isentropic turbine?

- A. Greater than
- B. Less than
- C. Same

Isentropic efficiency – turbine

Purpose: produce work

work done by real turbine, $\Phi > 0$

$$\eta_{isen,t} = \frac{\dot{W}_{real}}{\dot{W}_{ideal}} < 1, \quad \dot{W}_{real} < \dot{W}_{ideal}$$

work done by
a reversible, adiabatic turbine
 $\Delta S = 0$

$$\Rightarrow \eta_{isen,t} \approx 70 - 90\%$$

Isentropic efficiency – compressor

Goal : increase enthalpy through work input



Would a non-ideal compressor require greater than, less than, or the same amount of work to drive as an isentropic compressor?

- A. Greater than
- B. Less than
- C. Same

Isentropic efficiency – compressor

$$\eta_{\text{isen, comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} < 1, \quad \dot{W}_{\text{ideal}} < \dot{W}_{\text{real}}, \quad \eta_{\text{isen, comp}} \approx 75-85\%$$

$$\Rightarrow \dot{m} \Delta h = -\dot{W} \rightarrow \begin{cases} \dot{W}_{\text{real}} = -\dot{m} (h_2 - h_1) \\ \dot{W}_{\text{ideal}} = -\dot{m} (h_{2,s} - h_1) \end{cases} \rightarrow \text{State 2, } \underline{S_2} > S_1, P_2$$

where $h_{2,s}$ is the enthalpy $2s$ where $S_{2,s} = S_1$ (ideal), P_2

$\rightarrow$ turb: $h_{2,s} < h_2$ - less work extracted in real turb.

$\rightarrow$ comp: $h_{2,s} < h_2$ - less work input in ideal turbine